

Multi-Sampler Module Specification

Overview:

The Multi-Sampler is a lean, encapsulated module for extracting sparse samples from 2D image data. It combines two complementary strategies: **Random Sampling**: unbiased stochastic point selection across the image. **Wallis Sieve Sampling**: deterministic, structured halving-based point selection inspired by Wallis' product for π . This dual strategy yields both statistical representativeness and structured coverage, making it suitable for reconstruction pipelines, compression, inpainting, and feature extraction.

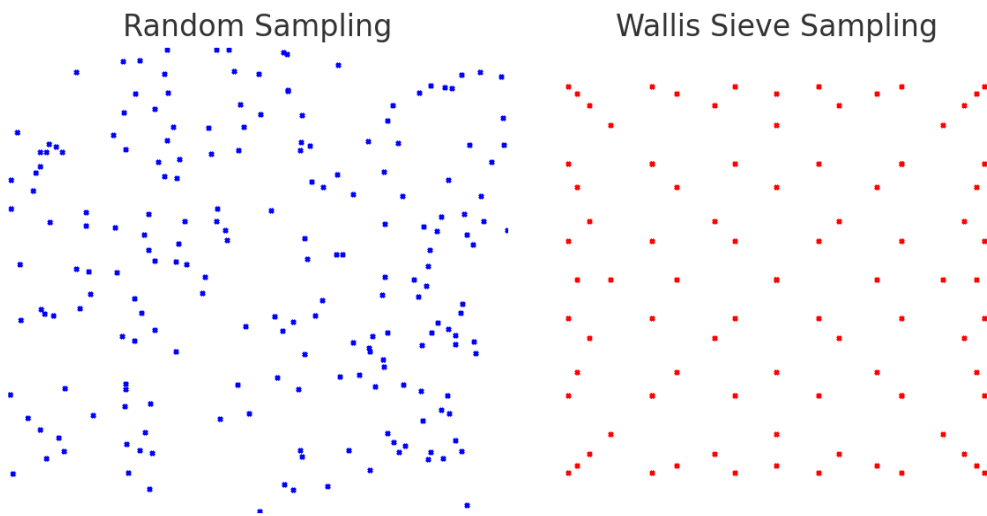
Design and Structure:

The module is lightweight, input-agnostic, and robust: Encapsulated in a single class: *MultiSampler*. Methods exposed for each sampling mode and combined sampling. Configurable parameters: number of samples, random seed, sieve depth. Output: list or array of (x, y) coordinates suitable for downstream use.

Methods and APIs:

`sample_random(n_points, seed=None)`: returns random sample coordinates. `sample_wallis(depth)`: returns Wallis sieve coordinates (grid halving method). `sample_dual(n_points, depth, ratio=0.5)`: returns mixed samples, weighted by ratio. `export(format='numpy' | 'list')`: output samples in requested format.

Sampling Strategy Comparison:



The Multi-Sampler can be applied to **sparse video frame extraction**, **compressed sensing**, and **reconstruction tasks**. Random sampling provides statistical diversity, while Wallis sieve ensures deterministic coverage. Together, they enable both efficient downstream reconstruction (TV inpainting, patch-based filling, U-Net reconstruction) and experimental use in fractal or structured compression pipelines.